

LM02 Understanding Business Cycles

1. Introduction.....	2
2. Overview of the Business Cycle.....	2
Phases of the Business Cycle	2
Leads and Lags in Business and Consumer Decision Making.....	4
Market Conditions and Investor Behavior.....	4
3. Credit Cycles	5
Applications of Credit Cycles.....	5
Consequences for Policy	5
4. Economic Indicators Over the Business Cycle.....	5
The Workforce and Company Costs	5
Fluctuations in Capital Spending.....	6
Fluctuations in Inventory Levels	7
Economic Indicators.....	7
Summary.....	10

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1. Introduction

This learning module covers:

- What is a business cycle; what are the different phases in a business cycle?
- Credit cycles and their relationship to business cycles
- How resource use, capital spending, and inventory levels vary over the business cycle
- Economic indicators that are useful in predicting the future of an economy.

2. Overview of the Business Cycle

The curriculum defines business cycle as: “Business cycles are a type of fluctuation found in the aggregate economic activity of nations that organize their work mainly in **business enterprises**: a cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions, contractions, and revivals which merge into the expansion phase of the next cycle; this sequence of events is **recurrent but not periodic**; in duration, business cycles vary from more than one year to 10 or 12 years.”

Some important points to be noted from the definition:

- Business cycles occur in economies where there are a large number of private companies, and not just agriculture economies.
- The economic activity shows a cyclical behavior between expansion and recession.
- They are pervasive, i.e., the cycle includes many economic activities and not just one sector. And the phases of expansion or contraction occur at the same time throughout the economy. For example, banking and real estate both may be in an expansion stage.
- They are recurrent but not periodic, i.e., the cycles repeat. To say they are not periodic means that the intensity and the duration differs. For instance, if an economic boom lasted for five years from 2002-07, it does not mean that the expansion phase will last for five years in the next cycle. Each cycle lasts about 1 to 12 years.

Phases of the Business Cycle

Types of Cycles

‘Classical cycle’ refers to fluctuations in the level of economic activity when measured by GDP in volume terms. It is rarely used in practice, because it does not easily allow us to distinguish between short-term fluctuations and long-term trends.

‘Growth cycle’ refers to fluctuations in economic activity around the long-term potential trend growth level. The focus is on how much actual economic activity is below or above trend growth in economic activity. The advantage of this method is that it divides economic activity into a part reflecting long-term trends and a part reflecting short-term fluctuations.

‘Growth rate cycle’ refers to fluctuations in the growth rate of economic activity. A growth

rate above potential growth rate reflects upswings, while a growth rate below potential growth rate reflects downswings. The advantage of this definition is that there is no need to estimate a long-run growth path first. Also, peaks and troughs can be recognized earlier than when using the other two definitions.

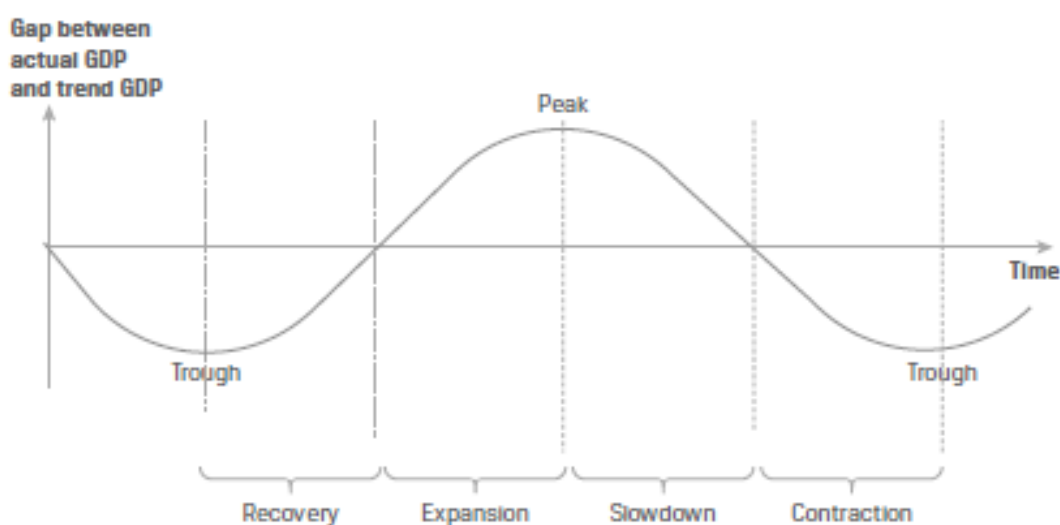
Four Phases of the Cycle

There are four phases of a business cycle:

1. Recovery: The economy is going through the trough of the cycle. Economic activity (which includes consumer and business spending) is below potential but is starting to increase.
2. Expansion: The recovery gathers momentum and economic activity rises above potential. The economy enters the so-called 'boom' phase.
3. Slowdown: The economy is going through the peak of the cycle. Economic activity is above potential but is starting to decrease.
4. Contraction: Economic activity falls below potential. The economy may experience a recession.

The four stages are illustrated below (this exhibit is reproduced from the curriculum):

Schematic of Business Cycle Phases



Some of the important characteristics of each phase in a business cycle are presented in Exhibit 5 from the curriculum.

Characteristics:

Phase	Recovery	Expansion	Slowdown	Contraction
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Description	Economy going through a trough. Negative output gap starts to narrow.	Economy enjoying an upswing. Positive output gap opens.	Economy going through a peak. Positive output gap starts to narrow.	Economy weakens and may go into a recession. Negative output gap opens.
Activity levels – consumers and businesses	Activity levels are below potential but start to increase.	Activity measures show above-average growth rates.	Activity measures are above average but decelerating. Moving to below-average rates of growth.	Activity measures are below potential. Growth is lower than normal.
Employment	Layoffs slow. Businesses rely on overtime before moving to hiring. Unemployment remains higher than average.	Businesses move from using overtime and temporary employees to hiring. Unemployment rate stabilizes and starts falling.	Businesses continue hiring but at a slower pace. Unemployment rate continues to fall but at decreasing rates.	Businesses first cut hours, eliminate overtime, and freeze hiring, followed by outright layoffs. Unemployment rate starts to rise.
Inflation	Inflation remains moderate.	Inflation picks up modestly	Inflation further accelerates.	Inflation decelerates but with a lag.

Leads and Lags in Business and Consumer Decision Making

The behavior of businesses and households often leads or lags the turning points of the business cycle. For example, at the start of an expansion phase, businesses rely on overtime and wait to hire new employees until they are certain that the economy is truly growing.

Market Conditions and Investor Behavior

Recovery phase: When markets expect the end of a recession and the start of an expansion phase, risky assets are repriced upwards. Investors start incorporating higher profit expectations into the prices of stocks and bonds. Equity markets typically hit a trough about 3-6 months before the economy hits the trough.

Expansion phase: The boom phase tests the limits of the economy. For example, companies may expand so much that they have difficulty finding qualified labor and will compete with

other companies by raising wages. Companies may also borrow a lot to fund capacity expansions. The government may have to step in to prevent the economy from overheating.

Slowdown phase: During the boom, the riskiest assets will usually have significant price increases. Whereas, safe assets such as government bonds may have lower prices and thus higher yields.

Contraction phase: During contraction, investors prefer safer assets such as government securities and defensive companies with stable positive cash flows.

3. Credit Cycles

Credit cycles describe the changing availability—and pricing—of credit. When the economy is strong the willingness of lenders to extend credit on favorable terms is high. Whereas, when the economy is weak lenders make credit less available and more expensive. This can result in decline of asset values and cause further economic weakness and higher defaults.

Applications of Credit Cycles

Credit cycles should be studied due to the importance of credit in the financing of construction and the purchase of property. The duration of recessions and recoveries are often shaped by linkages between business and credit cycles. Recessions accompanied by rapid fall in credit tend to be longer and deeper. Such situations can also lead to housing and equity price busts. Recoveries accompanied by rapid growth in credit tend to be stronger. They can also lead to a revival in house and equity prices.

Credit cycles are not always synchronized with the business cycle. They tend to be longer, deeper, and sharper than the business cycle.

Consequences for Policy

Investors pay attention to the stage in the credit cycle because:

- it helps them understand developments in the housing and construction markets
- it helps them assess the extent of business cycle phases
- it helps them better anticipate policy makers' actions

4. Economic Indicators Over the Business Cycle

The Workforce and Company Costs

Exhibit 7 from the curriculum depicts the pattern of hiring and employment through different phases of the business cycle.

Phase	Recovery	Expansion	Slowdown	Contraction
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Description of activity levels	Economy starts at trough and output below potential. Activity picks up, and gap starts to close.	Economy enjoying an upswing, with activity measures showing above-average growth rates.	Economy at peak. Activity above average but decelerating. The economy may experience shortages of factors of production as demand may exceed supply.	Economy goes into a contraction, (recession, if severe). Activity measures are below potential. Growth is lower than normal.
Employment	Layoffs slow. Businesses rely on overtime before moving to hiring. Unemployment remains higher than average.	Businesses move from using overtime and temporary employees to hiring. Unemployment rate stabilizes and starts falling.	Businesses continue hiring but at a slower pace. Unemployment rate continues to fall but at slowly decreasing rates.	Businesses first cut hours, eliminate overtime, and freeze hiring, followed by outright layoffs. Unemployment rate starts to rise.
	Levels of employment lag the cycle			

Fluctuations in Capital Spending

Exhibit 8 from the curriculum depicts the fluctuations in capital spending through different phases of the business cycle.

Phase	Recovery	Expansion	Slowdown	Contraction
Business conditions and expectations	Excess capacity during trough, low utilization, little need for capacity expansion. Interest rates tend to be low—supporting investment.	Companies enjoy favorable conditions. Capacity utilization increases from low levels. Over time, productive capacity may begin to limit ability to respond to demand. Growth in earnings and cash flow gives businesses the financial ability to increase investment spending.	Business conditions at peak, with healthy cash flow. Interest rates tend to be higher—aimed at reducing overheating and encouraging investment slowdown.	Companies experience fall in demand, profits, and cash flows.

Capital spending	Low but increasing as companies start to enjoy better conditions. Capex focus on efficiency rather than capacity. Upturn most pronounced in orders for light producer equipment. Typically, the orders initially reinstated are for equipment with a high rate of obsolescence, such as software, systems, and technological hardware.	Customer orders and capacity utilization increase. Companies start to focus on capacity expansion. The composition of the economy's capacity may not be optimal for the current structure of demand, necessitating spending on new types of equipment. Orders precede actual shipments, so orders for capital equipment are a widely watched indicator of the future direction of capital spending.	New orders intended to increase capacity may be an early indicator of the late stage of the expansion phase. Companies continue to place new orders as they operate at or near capacity.	New orders halted, and some existing orders canceled (no need to expand). Initial cutbacks may be sharp and exaggerate the economy's downturn. As the general cyclical bust matures, cutbacks in spending on heavy equipment further intensify the contraction. Maintenance scaled back.
Examples	Software, systems, and hardware (high rates of obsolescence) orders placed or re-instated first.	Heavy and complex equipment, warehouses, and factories. A company may need warehouse space in locations different from where existing facilities are.	Fiber-optic overinvestment in late 1990s that peaked with the "technology, media, telecoms bubble."	Technology and light equipment with short lead times get cut first. Cuts in construction and heavy equipment follow.

Fluctuations in Inventory Levels

Increase and decrease in inventory happens very rapidly, and has a major effect on economic growth despite the small size. $\frac{\text{Inventory}}{\text{Sales}}$ (I/S) ratio is an important indicator. Final sales numbers better reveal the reality of the economic situation than inventory numbers because the inventory may accumulate or companies may want to dispose obsolete inventory before starting production; it depends on the stage of the business cycle. Inventories tend to rise when the I/S ratio is low. During recovery, inventory will be less than sales and companies start production to increase inventory.

Exhibit 9 from the curriculum shows how production, sales, and inventories typically move through different phases of the business cycle.

Phase	Recovery	Expansion	Slowdown	Contraction
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Sales and production	Sales decline slows. Sales subsequently recover. Production upturn follows but lags behind sales growth. Over time, production approaches normal levels as excess inventories from the downturn are cleared.	Sales increase. Production rises fast to keep up with sales growth and to replenish inventories of finished products. This increases the demand for intermediate products. "Inventory rebuilding or restocking stage."	Sales slow faster than production; inventories increase. Economic slowdown leads to production cutbacks and order cancellations.	Businesses produce at rates below the sales volumes necessary to dispose of unwanted inventories.
Inventory-sales ratio	Begins to fall as sales recovery outpaces production.	Ratio stable.	Ratio increases. Signals weakening economy.	Ratio begins to fall back to normal.

Economic Indicators

Economic indicators are variables that are used to assess the state of the overall economy and for providing insights into future economic activity.

Types of Indicators

Economic indicators can be classified based on whether they lag, lead, or coincide with changes in an economy's growth.

- *Leading indicators* have turning points that tend to precede those of the business cycle. They help in forecasting the economy in the near term.
 - Ex: Weekly hours in manufacturing, S&P 500 return, private building permits.
- *Coincident indicators* have turning points that tend to coincide with those of the business cycle and are used to indicate the current phase of the business cycle.
 - Ex: Manufacturing activity, personal income, number of non-agricultural employees.
- *Lagging indicators* have turning points that tend to occur after those of the business cycle.
 - Ex: Bank prime lending rate, inventory-to-sales ratio, average duration of unemployment.

Composite Indicators

Composite indicators consist of a composite of different variables that all tend to move together. Different countries will have different composite indices. These indicators are based on empirical observations of an economy.

Leading Indicators

In the U.S., the composite leading indicator is called the Index of Leading Economic Indicators (LEI) that consists of 10 component parts. Exhibit 11 presents the 10 components used in the LEI.

1. Average weekly hours, manufacturing: Firms cut on overtime before a downturn and increase the overtime before hiring full-time workers during a recovery.
2. Average weekly initial claims for unemployment insurance.
3. Manufacturers' new orders for consumer goods and materials.
4. ISM new order index: The Institute of Supply Management (ISM) polls its members to build indexes of manufacturing orders, output, employment, pricing, and comparable gauges for services.
5. Manufacturers' new orders for non-defense capital goods excluding aircraft
6. Building permits for new private housing units.
7. S&P 500 stock index.
8. Leading credit index: Aggregates the information from six leading financial indicators, which reflect the strength of the financial system to endure stress.
9. Interest rate spread between 10-year Treasury yields and overnight borrowing rates: Spread is the difference between long-term yields and short-term yields. If the curve is upward sloping (a wider spread), then we expect short-term rates in the future to be high and more economic growth.
10. Average consumer expectations for business conditions

Using Economic Indicators

Leading indicators are particularly useful as they can help predict where the economy is likely to be in the near future. Coincident and lagging indicators can help confirm what the leading indicators are telling us.

Other Composite Leading Indicators

Apart from the LEI used in the US, there are several other composite leading indicators that are used across different countries.

Surveys

Composite indicators often make use of economic surveys. These surveys are usually conducted by central banks, research institutes, statistical offices, and trade associations on a monthly or quarterly basis. The surveys are conducted among either businesses, consumers, or experts. For example, among the 10 components of the LEI, the following three components are survey based:

- ISM new order index
- Manufacturers' new orders for non-defense capital goods excluding aircraft
- Average consumer expectations for business conditions

The Use of Big Data in Economic Indicators

In recent years, due to the vast increase in the amount of information and the number of

variables that go into composite indicators, statistical techniques such as 'principal component analysis' are frequently used while constructing indexes using economic indicators.

Nowcasting

Policy makers and market practitioners have started monitoring economic and financial variables such as internet searches and electronic payment data in real-time. This allows them to continuously assess current conditions and produce a nowcast.

Nowcasting produces an estimate of the current state of the economy. The advantage of this method is that it helps overcome delays associated with the release of actual measures. For example, measures such as GDP are only published with delay, after the end of the time period under consideration.

GDPNow

GDPNow is a nowcast published by the Federal Reserve Bank of Atlanta. According to the Atlanta Fed, "GDPNow" is "best viewed as a running estimate of real GDP growth based on available data for the current measured quarter."

Summary

LO: Describe the business cycle and its phases.

There are four phases of a business cycle:

1. Recovery: The economy is going through the trough of the cycle. Economic activity (which includes consumer and business spending) is below potential but is starting to increase.
2. Expansion: The recovery gathers momentum and economic activity rises above potential. The economy enters the so-called 'boom' phase.
3. Slowdown: The economy is going through the peak of the cycle. Economic activity is above potential but is starting to decrease.
4. Contraction: Economic activity falls below potential. The economy may experience a recession.

LO: Describe credit cycles.

Credit cycles describe the changing availability—and pricing—of credit. When the economy is strong the willingness of lenders to extend credit on favorable terms is high. Whereas, when the economy is weak lenders make credit less available and more expensive. This can result in decline of asset values and cause further economic weakness and higher defaults.

LO: Describe how resource use, consumer and business activity, housing sector activity, and external trade sector activity vary over the business cycle and describe their measurement using economic indicators.

Recovery:

- GDP growth rate changes from negative to positive.
- High unemployment rate and a moderate or declining inflation.
- Increasing production to meet the pickup in sales with more flexible methods like overtime or increasing utilization levels.

Expansion:

- GDP growth rate increases.
- Reduction in unemployment rate as hiring rises.
- Inflation may begin to rise.
- Increasing production needs are met with investments and labor force additions.

Slowdown:

- GDP growth rate decreases.
- Unemployment rate decreases, but firms cut back on hiring.
- Business and consumer confidence declines, slowing the growth rates in investments and consumer spending.
- Inflation rate increases.

Contraction:

- GDP growth rate is declining.
- Unemployment rate increases as firms cut back on production.
- Inflation decreases with a lag.
- Decline in consumer and business confidence lowers the investment and consumer spending.

The three types of indicators are:

- Leading indicators:
 - Indicators that change direction before the peaks and troughs of business cycles.
 - Examples: Stock prices, index of consumer expectations etc.
- Coincident indicators:
 - Indicators that change direction at roughly the same time as the peaks and troughs of business cycles.
 - Examples: Industrial production, manufacturing and trade sales etc.
- Lagging indicators:
 - Indicators that do not change direction until the expansions and contractions are already in progress.
 - Examples: inventory-sales ratio, unemployment rate, labor cost per unit of output, etc.